

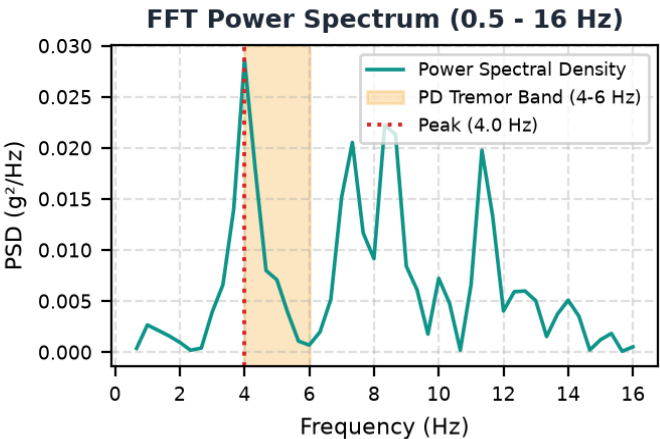
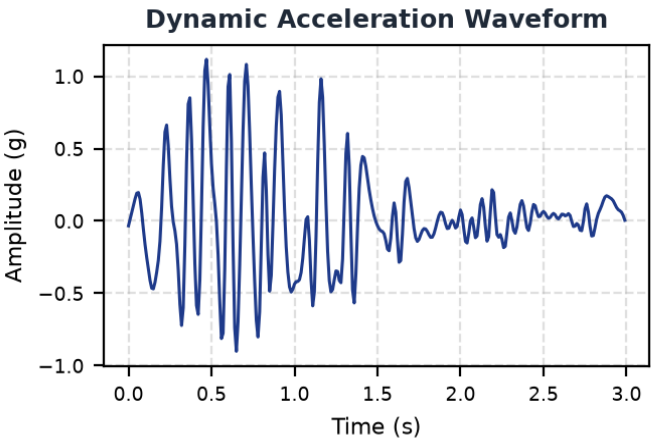
NEUROTRACK — Tremor Assessment & Telemetry Report

Report ID: NT-1788584498
Date: 2026-09-05 10:31:38

Patient Identifier:	LIVE_COM4	Data Origin:	■ Live Physical Hardware (ESP32/COM4)
Sensor Device:	Physical ESP32 + MPU6050 (COM4)	Sampling Rate:	50.0 Hz (Nyquist: 25.0 Hz)
Assessment Duration:	6.0 seconds	Analysis Window:	3.0 sec (50% sliding overlap)

AI Pattern Classification	Tremor Severity Index	Clinical Severity Grade
HEALTHY Confidence: 99.0% (PD Match: 0.0%)	0.0 / 100	Minimal / Negligible Physiological baseline or voluntary movement. No rhythmic 4-6 Hz tremor detected.

Biomechanical Time-Series & Spectral Isolation



Extracted Digital Biomarkers (Window Metrics)

Biomarker	Observed Value	Reference Band / Diagnostic Context
Dominant Frequency	4.00 Hz	Parkinsonian Resting Tremor: 4.0 - 6.0 Hz
Tremor Band Power Ratio	21.6%	Ratio of power in 4-6 Hz band to broad spectrum (>40% indicates localized tremor)
Dynamic Jerk RMS	19.06 g/s	Time-derivative of acceleration measuring abrupt oscillatory changes
Spectral Shannon Entropy	0.882	Low (<0.45) indicates rhythmic periodicity; High (>0.70) indicates random/voluntary noise

Physiological Interpretation & Explainability

Dominant frequency (4.0 Hz) falls precisely within the 4.0 - 6.0 Hz cardinal resting tremor band associated with Parkinsonian motor signs. Moderate tremor-band power ratio (21.6%) indicates intermittent or mild periodicity. Spectral entropy (0.88) reflects broad-spectrum kinetic noise or non-rhythmic movement. Dynamic amplitude RMS (0.352 g) and jerk (19.1 g/s) signify substantial biomechanical displacement and rapid oscillatory acceleration.

NOTICE & MEDICAL DISCLAIMER: NeuroTrack is an experimental screening and longitudinal monitoring aid, NOT a diagnostic or medical treatment device. The metrics, classifications, and severity scores provided herein reflect biomechanical time-series observations and must be interpreted solely by a qualified neurologist or physician in clinical context. Never initiate, alter, or discontinue any medication based on this report.